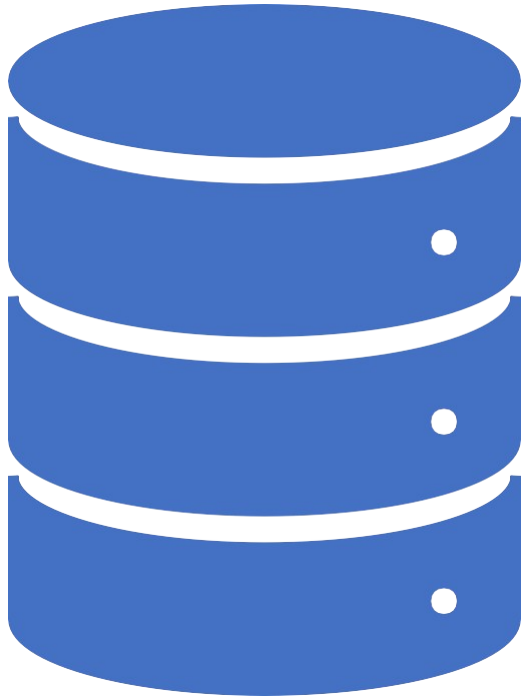
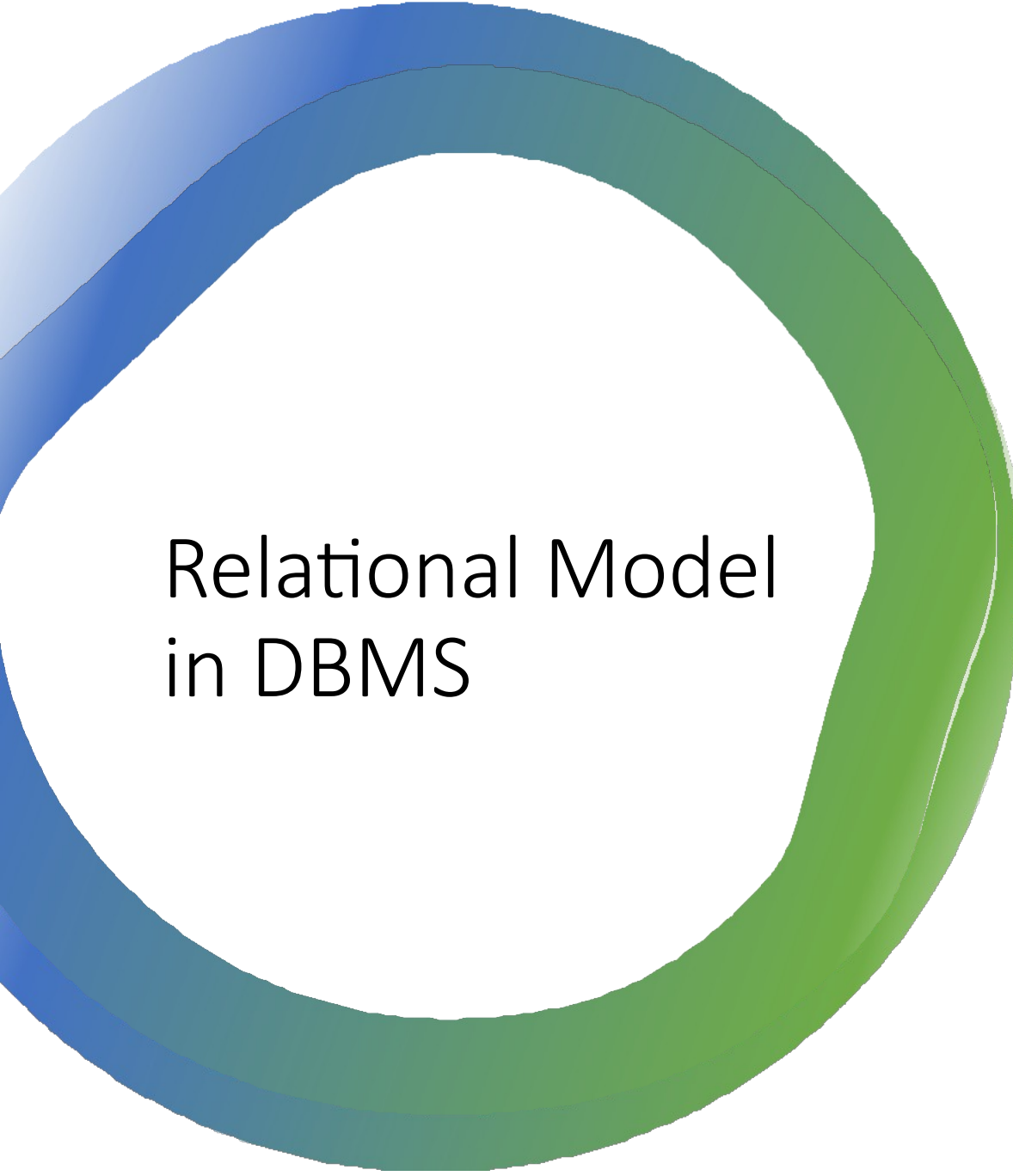


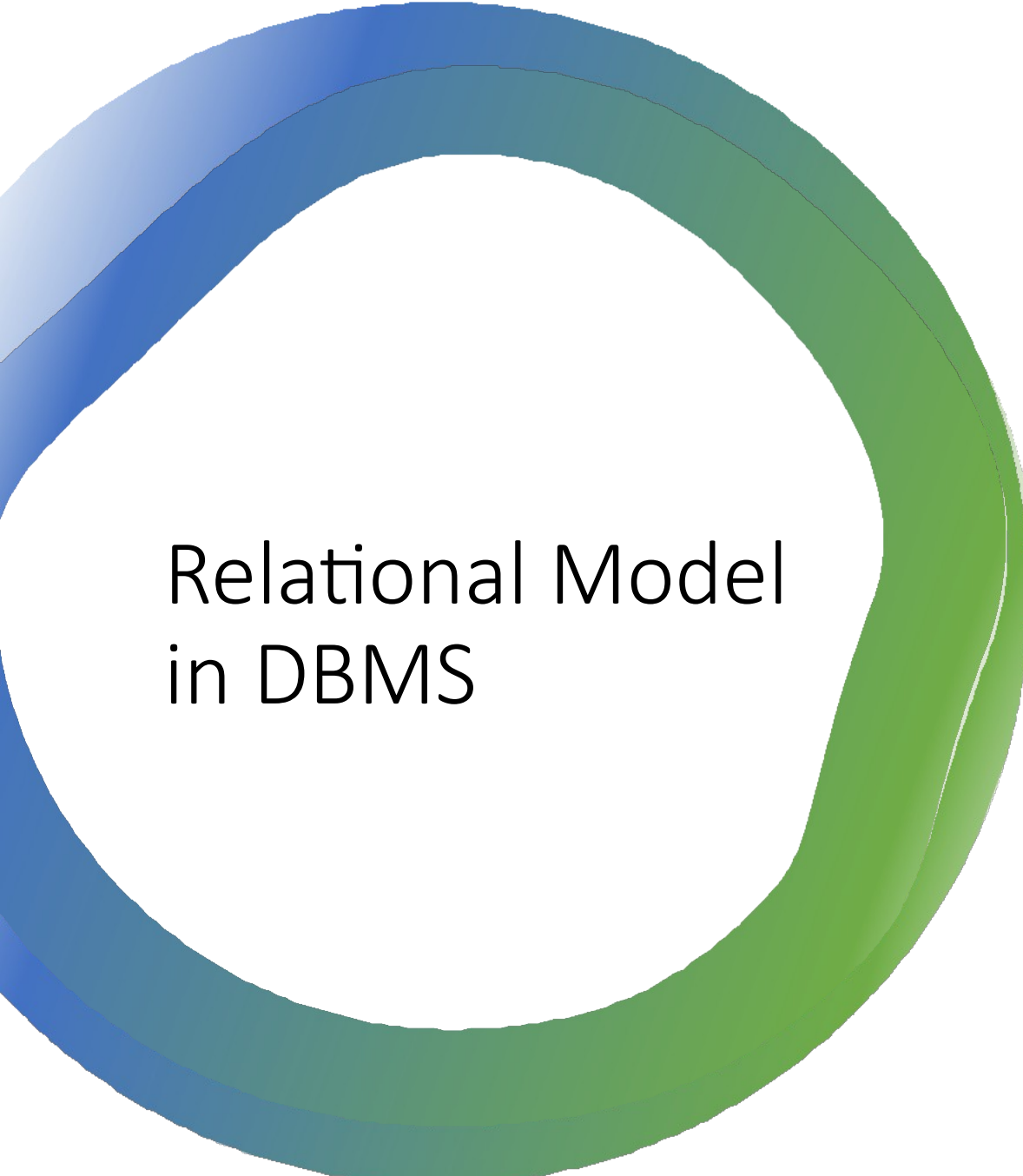


Database System
[*Relational Model in DBMS*]



Relational Model in DBMS

- Relational model can represent as a table with columns and rows.
 - Each row is known as a ***tuple***.
 - Each table of the column has a name or ***attribute***.



Relational Model in DBMS

Domain:

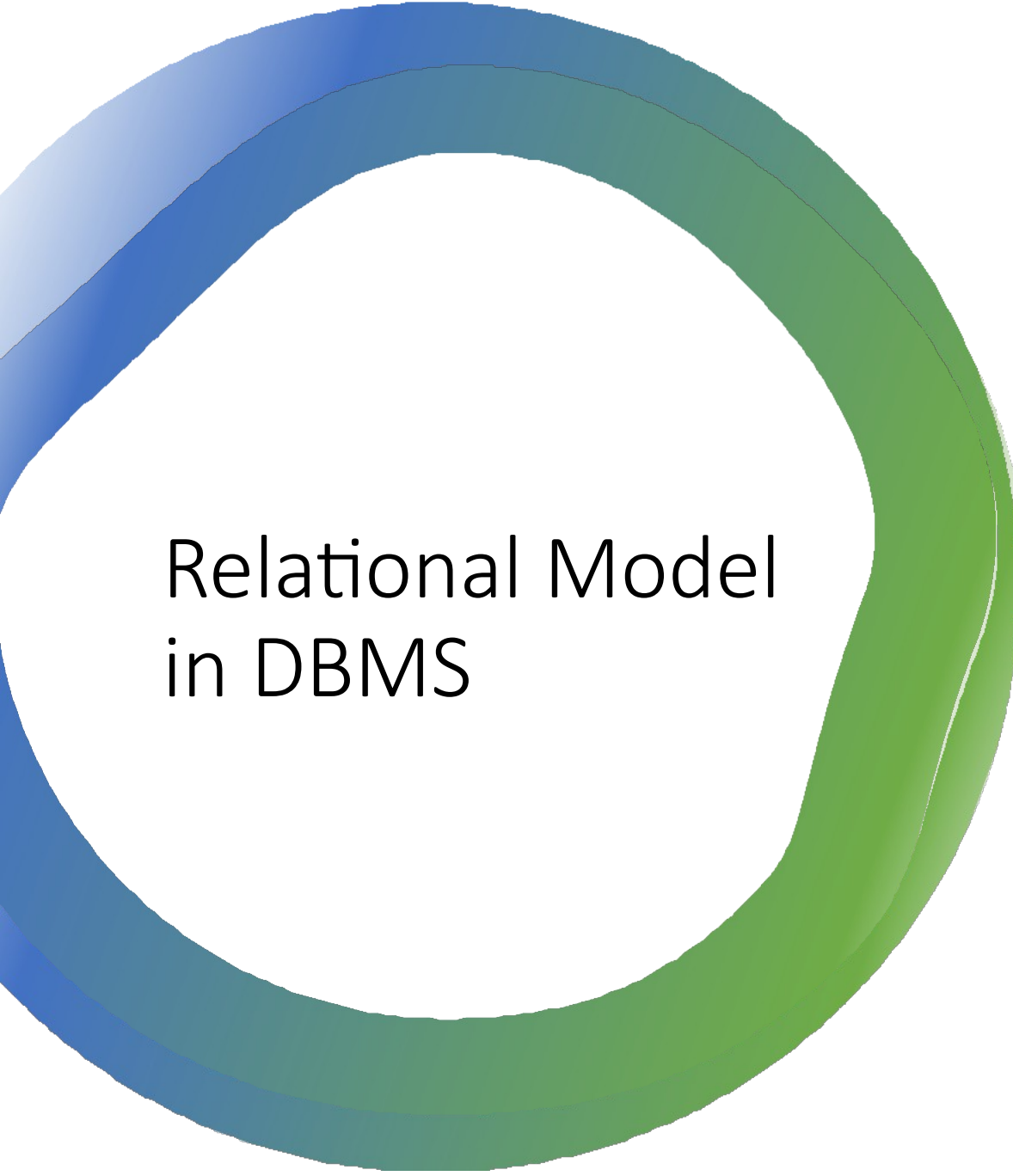
- It contains a set of atomic values that an attribute can take.

Attribute:

- It contains the name of a column in a particular table.
- Each attribute A_i must have a domain, $dom(A_i)$

Relational instance:

- In the relational database system, the relational instance is represented by a finite set of tuples.
- Relation instances do not have duplicate tuples.



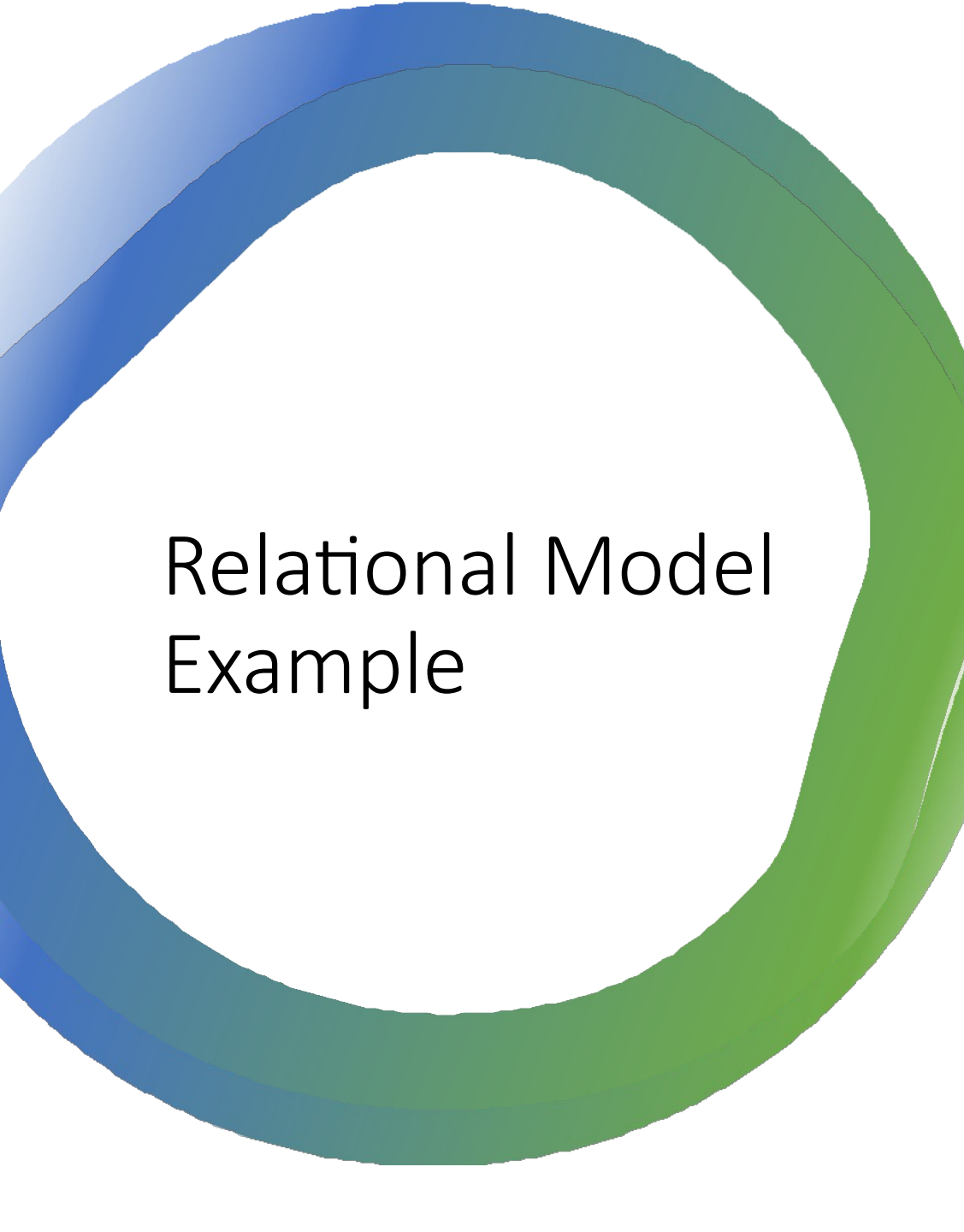
Relational Model in DBMS

Relational schema:

- A relational schema contains the name of the relation and name of all columns or attributes.

Relational key:

- In the relational key, each row has one or more attributes.
- It can identify the row in the relation uniquely.



Relational Model Example

STUDENT Relation

NAME	ROLL_NO	PHONE_NO	ADDRESS	AGE
Ram	14795	7305758992	Noida	24
Shyam	12839	9026288936	Delhi	35
Laxman	33289	8583287182	Gurugram	20
Mahesh	27857	7086819134	Ghaziabad	27
Ganesh	17282	9028 9i3988	Delhi	40

- In the given table, NAME, ROLL_NO, PHONE_NO, ADDRESS, and AGE are the attributes.
- The instance of schema STUDENT has 5 tuples.
- t3 = <Laxman, 33289, 8583287182, Gurugram, 20>



Properties of Relations

Name of the relation is distinct from all other relations.

Each relation cell contains exactly one atomic (single) value

Each attribute contains a distinct name

Attribute domain has no significance

tuple has no duplicate value

Order of tuple can have a different sequence